Group Project Phase 1:

Literature Review and Exploratory Data Analysis

Group 17

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Literature Review

Introduction

Effective crime prevention is fundamental to societal well-being and the advancement of social justice (Welsh & Piza, 2024), necessitating robust predictive policing models that facilitate preemptive police interventions for high-risk locations and individuals. Prediction accuracy is particularly important in this context since an underperforming model not only results in inefficient allocation of policing resources but also brings about ethical concerns (Griffard, 2019). However, as criminal datasets expand in scale and complexity, traditional manual analysis or simple statistical methods have become insufficient. This has driven the widespread application of machine learning algorithms to improve the accuracy of predictive policing (Meijer & Wessels, 2019). This review will evaluate the systems for predictive policing using past academic work and provide a comprehensive analysis of the various machine learning approaches applied.

Previous and Existing Systems In Literature

1. Evolution of Predictive Policing Systems

Historically, predictive policing relied heavily on traditional models. Early crime hotspot analysis primarily utilized point-pattern analysis (Chainey et al., 2002) and historical aggregated crime counts to identify spatial associations (Anselin, 1995). However, these approaches were frequently critiqued for being retrospective, as they operated on the unrealistic assumption that future criminal activity would strictly mirror past incidents.

In response, contemporary research has shifted towards prospective modelling. Empirical evidence suggests that the Risk Terrain Modeling (RTM), one of the most widely used models by law enforcement agencies to date (Hart, 2020), significantly outperforms retrospective methods in predicting crime hotspots (Caplan et al., 2011). Nevertheless, RTM remains constrained by user's subjectivity, as including or excluding specific risk-factor layers rests on the user's discretion, potentially leading to bias and impairing predictive reliability.

2. The Advent of Machine Learning in Criminology

The transition towards machine learning has further enhanced predictive accuracy. In the extant literature, Random Forest (RF) is frequently reported as a superior performer for crime hotspot prediction. For instance, Raza and Victor (2021) observed that employing RF to forecast the number of crimes to occur in experimental regions achieved 91% accuracy, significantly outperforming Logistic Regression (LR) (74%) and Support Vector Machines (SVM) (48%). Similar findings were documented by Kshatri et al. (2021), where RF reached 97% accuracy compared to 67% for Naïve Bayes (NB) model. However, it appears that this dominance does not extend to all categories of predictive policing. Suh et al. (2025) found that compared to RF, LR and elastic net regression (ENR), Gradient Boosting Machines (GBM) were proven to be optimal for predicting potential offenders' behavior. And surprisingly, k-Nearest Neighbors (KNN), despite its relative simplicity, has shown highest accuracy in victimization prediction, achieving an F1 score of 98% (Masmoudi et al., 2025).

Modeling Approach

Previous research on crime analysis often combines unsupervised learning, supervised learning, and regression methods to conduct the study of crime data, since they are usually very complex and contain different types of information.

K-means is one of the many clustering algorithms that are applied in analyzing crime incidents

based on locations to identify both spatial and temporal patterns. Hajela et al. (2020) utilized K-means to segment the dataset before feeding data to subsequent crime hotspot prediction algorithms and their research results showed that incorporating K-means clustering significantly enhanced prediction accuracy compared to the baseline model.

Supervised classification algorithms, including NB, RF, SVM, KNN and GBM are also commonly used to predict crime categories or crime occurrence. To enhance the predictive performance of these models, Grid Search (GS) is widely utilized as an optimization technique (Sagala, 2022). For instance, Suh et al. (2025) integrated 10-fold cross-validation into GS and found the optimal combination of hyperparameters that maximized prediction accuracy. Similarly, Raza and Victor (2021) leveraged GS to fine-tune their decision tree models and determined the optimal split based on entropy and Gini's criterion.

Apart from machine learning classification models, regression models like linear and multiple linear regression were often used to analyze the connection between dependent variables like crime rates, and independent variables like the victim's age, gender, incident's location and time (Sridharan et al., 2023; Mandalapu et al., 2023). For temporal forecasting, Auto Regressive Integrated Moving Average (ARIMA) is best suited to carry out predictive analysis for long periods, Random Forest Regressor excels at handling nonlinear relationships, while XGBoost can greatly help in improving precision and managing complex interactions (Safat et al., 2021).

Before applying these models, the data is first cleaned by handling missing values (such as with Grey-Based Weighted K-Nearest Neighbor) (Sun et al., 2014) and reducing noise, as poor data quality can lower classification accuracy and reliability (Safat et al., 2021).

Feature Engineering Techniques

The efficacy of the model, especially machine learning algorithms, is fundamentally dependent on feature engineering (Duboue, 2020). Some preprocessing procedures are universal standards, such as mitigating missing values either through removal of all records with missing data to improve computational efficiency (Kshatri et al., 2021) or via statistical imputation like replacing empty fields with mean or median (Masmoudi et al., 2025). Nevertheless, some other preprocessing requirements often vary significantly by model. For example, Duboue (2020) notes that normalization is crucial for SVM but optional for tree-based algorithms which are scale-invariant.

Additionally, reducing features is central to feature engineering (Duboue, 2020). As the findings of Suh et al.(2025) suggested, including additional unnecessary predictors can potentially impede model performance as a result of increased noise in the dataset or model overfitting. While RF is often used for feature selection due to its ability to rank variable importance (Masmoudi et al., 2025), it tends to overvalue variables with numerous unique levels, introducing cardinality bias. Alternatively, in the study conducted by Elluri et al. (2019), they leveraged Forward Stepwise Selection, which starts with an empty model and adds the most impactful predictors one by one, as well as Backward Stepwise Selection that begins with a full model and removes the least significant variables sequentially. While these methods are more computationally efficient than RF and offer higher interpretability, they rest on linear assumptions which can be limiting. Hence, a more robust approach, where computational resources allow, involves combining statistical approaches and RF to ensure non-linear interactions are captured and no redundant high-cardinality features are included.

Evaluation Metrics and Effectiveness

Beyond model construction and feature engineering, evaluating model effectiveness is equally critical in crime prediction research. Since feature engineering decisions may influence model performance, proper evaluation strategies, such as cross-validation, are essential to detect overfitting caused by excessive or irrelevant features (Suh et al. ,2025).

Previous studies evaluated the performance of crime prediction models using commonly adopted classification metrics, including accuracy, precision, recall, and F1-score. Accuracy reflects overall prediction performance, while precision and recall evaluate different types of classification errors, particularly false positives and false negatives. For multi-class or imbalanced crime data, F1-score is commonly used because it balances both precision and recall, as reported by Safat et al. (2021) and Sridharan et al. (2023).

In addition to accuracy-based metrics, ROC-AUC is used to evaluate how well classification models perform across different thresholds. Hajela et al. (2020) used this metric to evaluate how well their model separated crime categories. Similarly, a recent offender prediction research carried out by Suh et al. (2025) adopted AUC as a primary evaluation metric, accompanied by sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) to provide a more comprehensive assessment of classification performance. These measures are important in risk prediction studies, as they help evaluate how accurately the model identifies high-risk individuals.

In crime forecasting, metrics such as the Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) are normally employed to evaluate the performance of models such as ARIMA. As demonstrated by Safat et al. (2021), RMSE pays more attention to larger prediction errors, while MAE shows the average difference between predicted and actual values. These metrics assess the deviation between predicted and actual crime rates, making them suitable for measuring the performance of trend forecasting.

Conclusion

A review of previous literature reveals a progressive shift from traditional models to machine learning algorithms in predictive policing. Although tree-based algorithms tend to deliver optimal results in most cases, their efficacy remains contingent on the specific subcategory of predictive policing that the predictive task falls into. Moreover, the selection of evaluation metrics should align with the predictive objective to ensure that model performance is assessed appropriately. Feature engineering and data preprocessing also play crucial roles in safeguarding model robustness and generality. A critical limitation of prior studies lies in the failure to disaggregate the results by crime type. Understanding the types of crime expected in identified high-risk areas or the offenses that a potential offender or victim may be involved into, can help law enforcement agencies design more targeted interventions.

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Exploratory Data Analysis Report

Dataset Overview and Data Quality (Figure 1)

The dataset comprises 2,519,504 Chicago crime records (2015–2024) with 22 variables covering temporal, categorical, and spatial dimensions. After rigorous quality assessment and data cleaning, 2,425,599 records (96.3%) were retained with valid geographic coordinates and essential crime classifications, and the dataset was reduced to 13 key variables. This high retention rate and comprehensive coverage provide a robust foundation for detailed temporal and spatial pattern analyses.

Temporal Crime Patterns (Figure 2.1-Figure 2.3)

Temporal crime analysis reveals pronounced cyclical patterns. From 2015 to 2024, total crime declined by approximately 33%, decreasing from about 300,000 to 200,000 incidents, indicating a sustained downward trend without identifiable causal attribution. Seasonally, crime incidents peaked in summer (June–August). Diurnal patterns differed by crime type: property crimes mainly concentrated in daytime hours (12:00–18:00), whereas violent crimes peaked from late afternoon through late night (15:00–02:00). Weekend periods exhibited elevated late-night crime (20:00–02:00) relative to weekdays, underscoring the role of social and behavioral factors in driving crime incidents.

Crime Density Across Chicago (Figure 3)

The crime heatmap identifies a distinct vertical corridor extending southward from central Chicago, characterized by several high-density hotspots. The concentration within the urban core confirms that incidents are non-randomly clustered in the areas of high human activities and population density. A notable decrease in density is visible moving toward western and suburban edges, showing a clear spatial gradient in Chicago's crime distribution map.

Top Community Areas by Crime Incidents (Figure 4)

The distribution of crime incidents across community areas shows clear spatial concentration. Community Area 25 accounts for the highest number of reported incidents (140,222), followed by Community Area 8 (104,705). The remaining top-ranked areas record between approximately 56,000 and 88,000 incidents. This uneven distribution demonstrates that crime is not uniformly spread across neighborhoods but is more concentrated in specific communities.

Crime Type Analysis and Arrest Patterns (Figure 5)

Crime types exhibit substantial variation in both incidence and arrest rates. The five most frequent offenses previously identified, including theft, battery, criminal damage, assault, and deceptive practice, account for over 60% of all crime incidents. Yet arrest rates differ markedly, with theft, criminal damage and deceptive practice typically below 10%, whereas narcotics, weapons-related and public-peace-related offenses exceeding 50%. Predictive policing can bridge this gap by identifying specific times and locations where these low-arrest crimes are most likely to occur, allowing for preventative patrols in these areas.

Domestic Crime Patterns (Figure 6)

Domestic crime analysis indicates a strong concentration in assault-related offenses, with domestic battery and assault comprising the majority of domestic incidents. Approximately 20% of all crimes are domestic on average, though this share varies markedly by crime type. Offences involving children exhibit a domestic proportion near 80%, and battery above 50%, while property crimes

(e.g. theft, burglary and robbery) and narcotics offenses show minimal domestic involvement. The close association between crime type and domestic status suggests potential redundancy in predictive models, although their interaction, especially for assault-related crimes may retain predictive value.

Venue-Crime Type Correlations (Figure 7)

Cross-tabulation of crime type and venue reveals pronounced clustering. Streets, apartments and residences dominate overall. More specifically, theft, motor vehicle theft and criminal damage occur mostly on streets while battery is most frequent in apartments and residences. The weak association (Cramér's V：0.1962) indicates that venue may not be a strong contextual predictor complementary to crime type. Nevertheless, given the high cardinality of venue categories, Cramér's V can sometimes be artificially low if the data is spread too thin across many categories. Consequently, aggregating venue categories may preserve the variable as a meaningful feature.

Temporal-Spatial Interactions (Figure 8)

The spatial distribution of crime incidents remains consistent across years, with different time periods showing overlapping geographic patterns. This suggests that crime locations are stable over time rather than shifting to new areas.

Key Findings and Implications

Our findings indicate that spatial and temporal factors are robust predictors for the predictive policing model, given the consistent geographic concentration of crime hotspots and the distinct cyclical patterns observed throughout the past decade. Specifically, the vertical corridor concentration and late-night peaks for violent crimes provide reliable signals for predictive policing. Additionally, the high density of crime incidents within five specific crime types and the 50% domestic rate for battery reveal predictable behavioural patterns, making crime type and domestic status critical features for the model as well. While the association between crime type and venue is weak, venue remains a potential predictor if handled properly.

Conclusion

Public safety planning should focus first on stable hotspot locations and then adjust patrol times for busy periods. The high number but low clearance of property crimes suggests that the efficiency in allocation of policing resources needs to be improved. Domestic crimes may require support and intervention beyond normal patrol work. Future prediction models should group similar locations and focus on the Location–Type–Time relationship to support more proactive monitoring.

Dashboard link: https://it5006-app-dashboard-rjcapp4mbjmtqtvbvveygtl.streamlit.app/
(Please kindly note that the live link dashboard utilizes a data subset (2022-2024). This is because Streamlit Cloud requires the source data to be hosted on Github, whereas Github's file size limit (100MB) prevents the complete dataset from being uploaded. Please refer to the source code for the full-scale version (2015-2024) in Github via the following link, and it can be run through steps in readme:
https://github.com/LisazhenYANG/IT5006-Group-Project/blob/main/dashboard/InteractiveDashboardForLive.py)

GitHub repository link: https://github.com/LisazhenYANG/IT5006-Group-Project/tree/main

Appendix

```
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     2   date                  str
     3   block                 str
     4   iucr                  str
     5   primary_type          str
     6   description           str
     7   location_description   str
     8   arrest                bool
     9   domestic              bool
     10  beat                  int64
     11  district              float64
     12  ward                  float64
     13  community_area        float64
     14  fbi_code              str
     15  year                  int64
     16  updated_on            str
     17  x_coordinate          float64
     18  y_coordinate          float64
     19  latitude              float64
     20  longitude             float64
     21  location              str
    dtypes: bool(2), float64(7), int64(3), str(10)
    memory usage: 389.3 MB
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 1   primary_type          str
 2   description           str
 3   location_description  str
 4   arrest                bool
 5   domestic              bool
 6   beat                  int64
 7   district              float64
 8   ward                  float64
 9   community_area        float64
...
 11  latitude              float64
 12  longitude             float64
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Figure 1

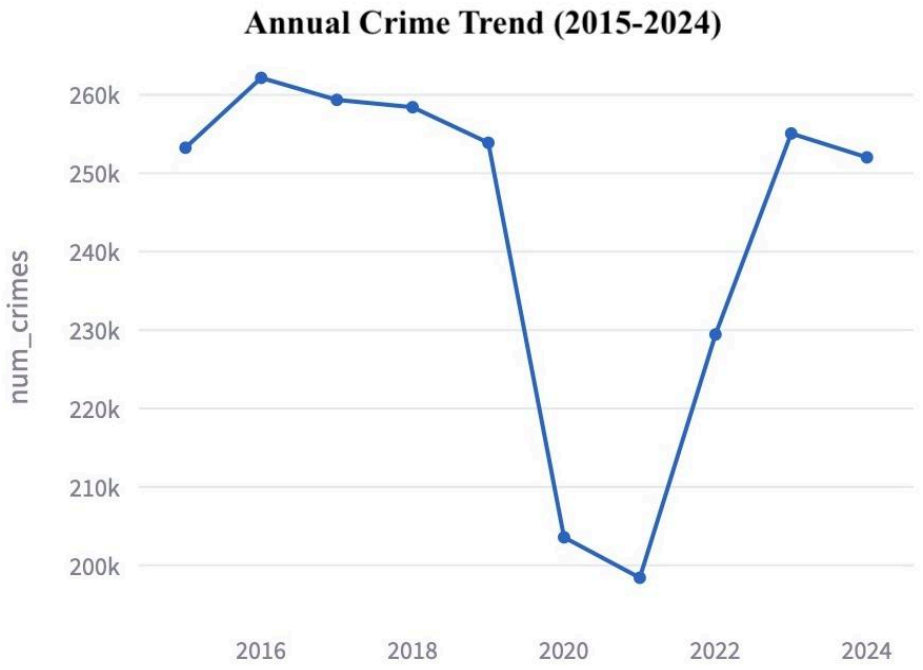
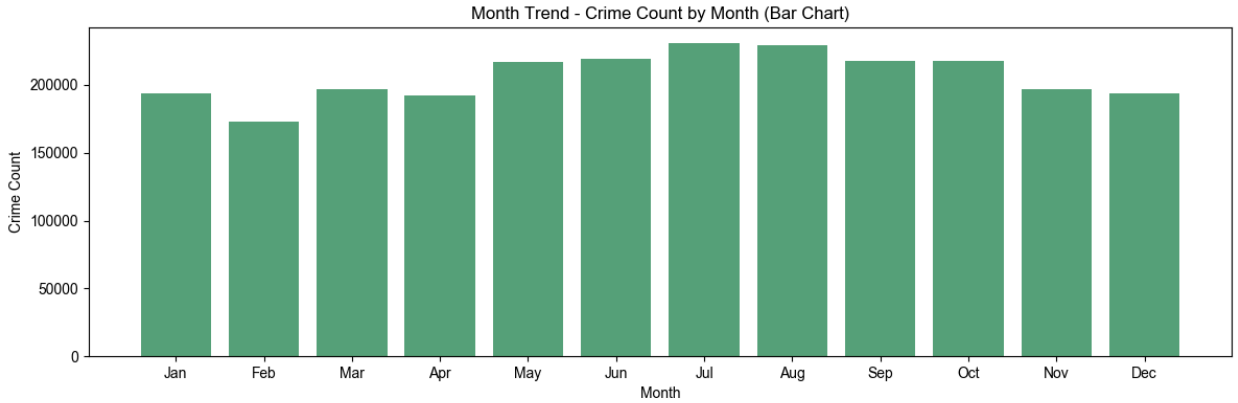


Figure 2.1



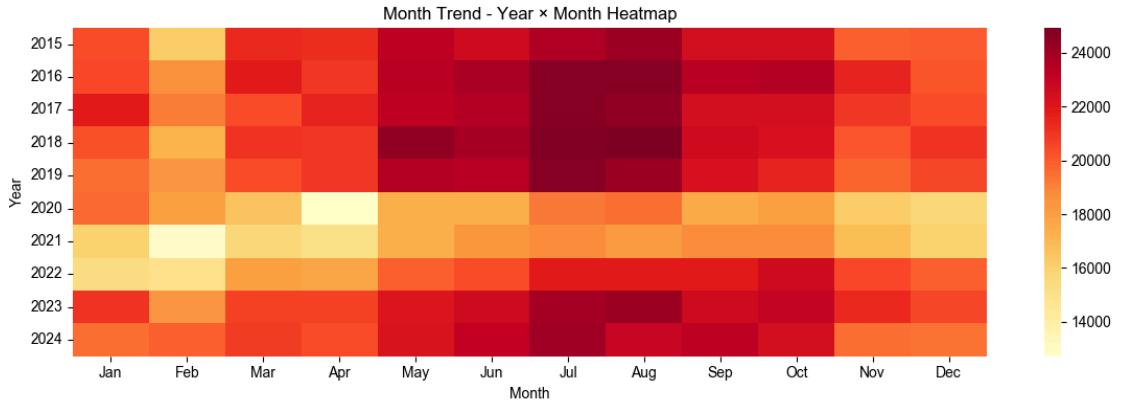
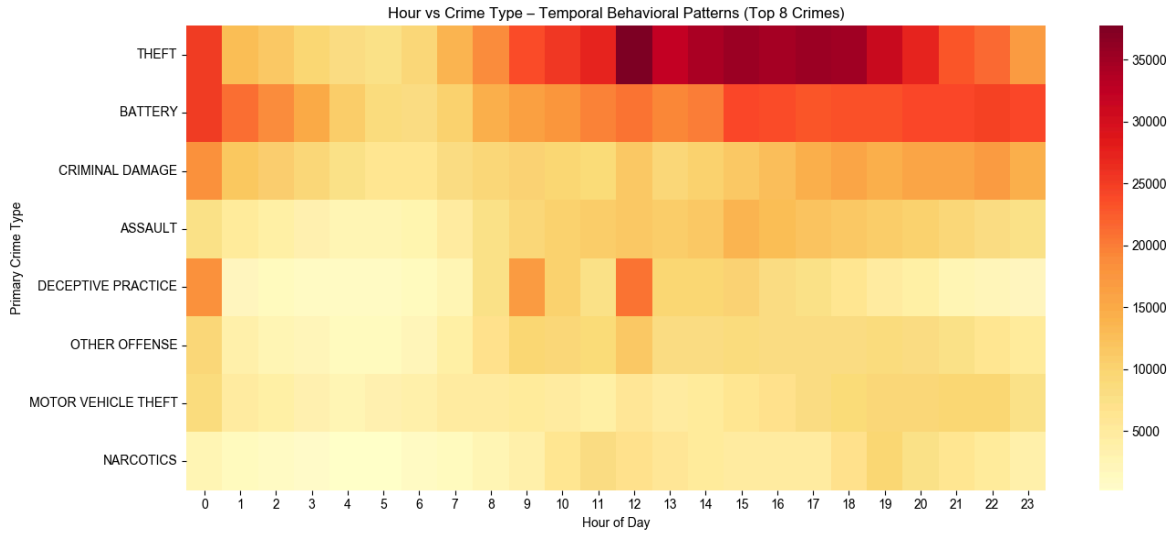


Figure2.2



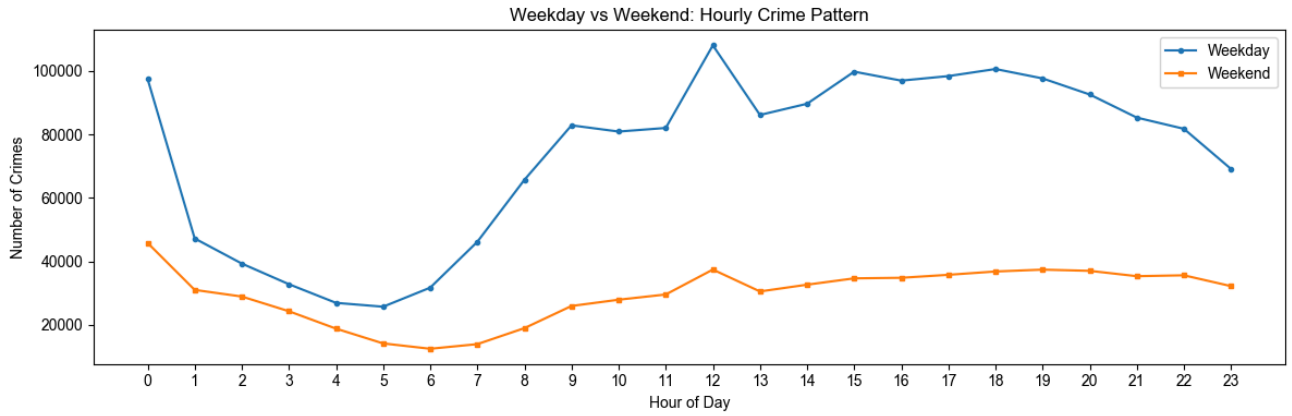


Figure 2.3

Crime heatmap (2015-2024)

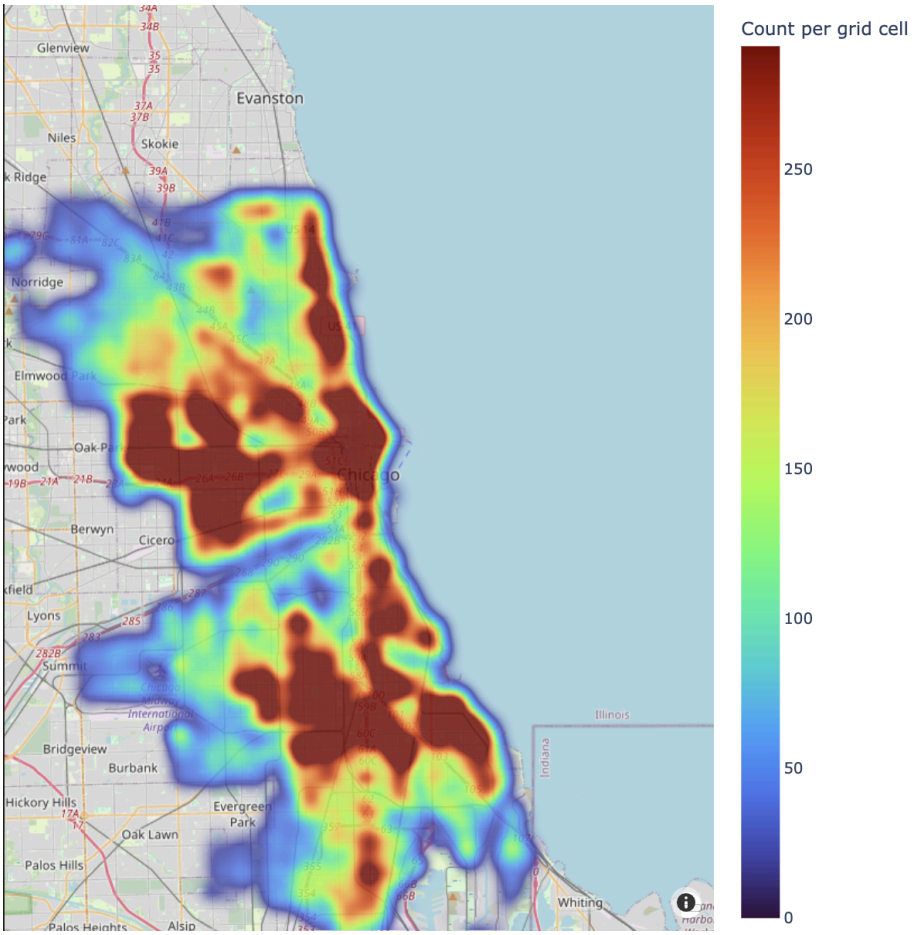


Figure 3

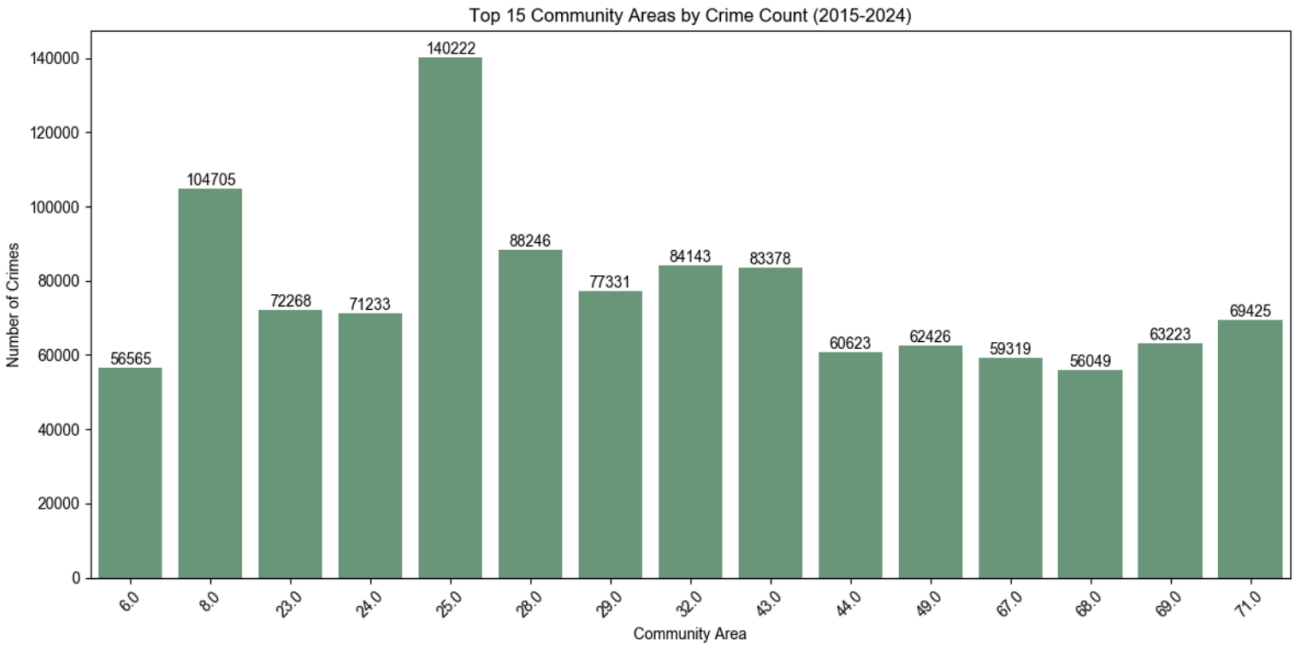


Figure 4

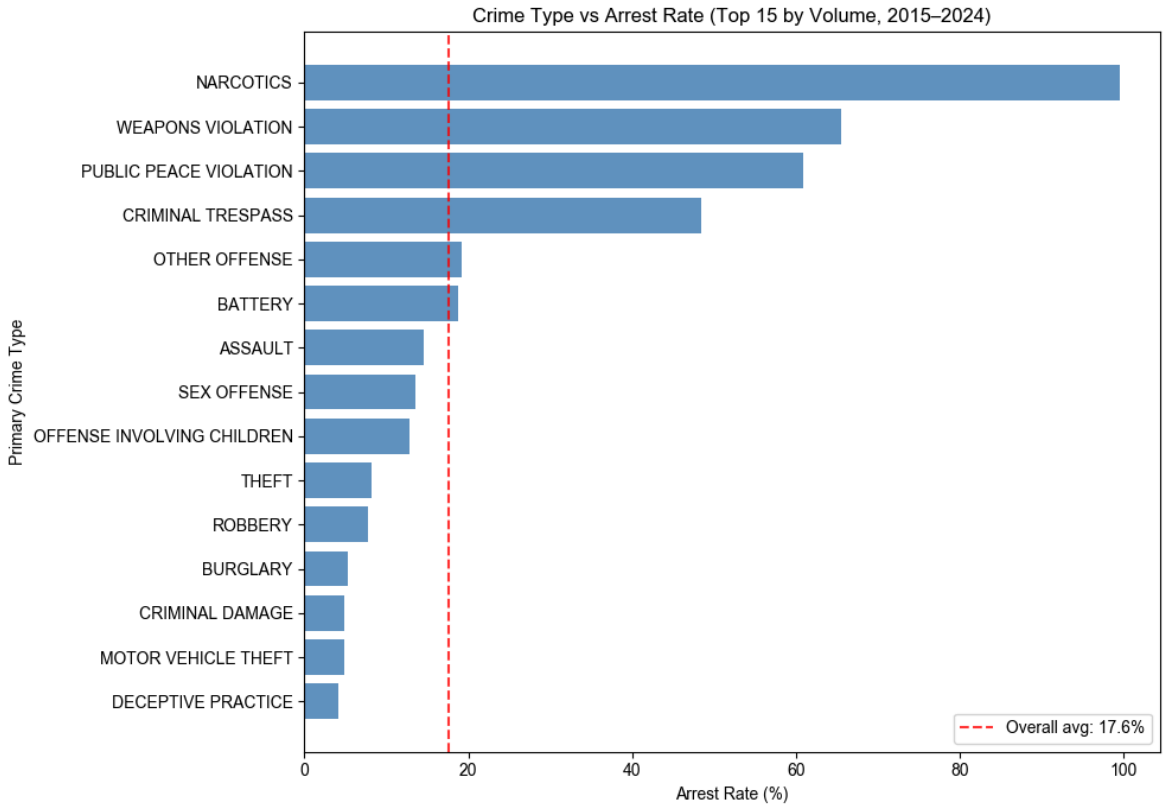


Figure 5

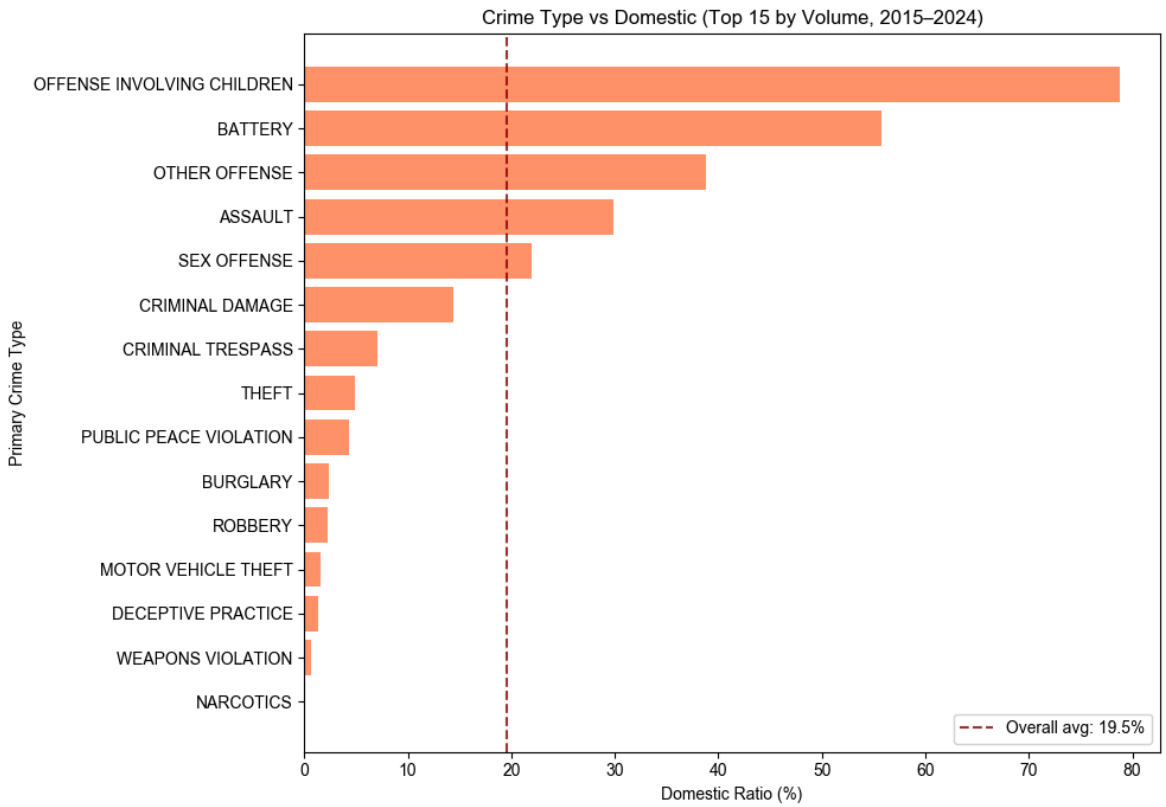


Figure 6

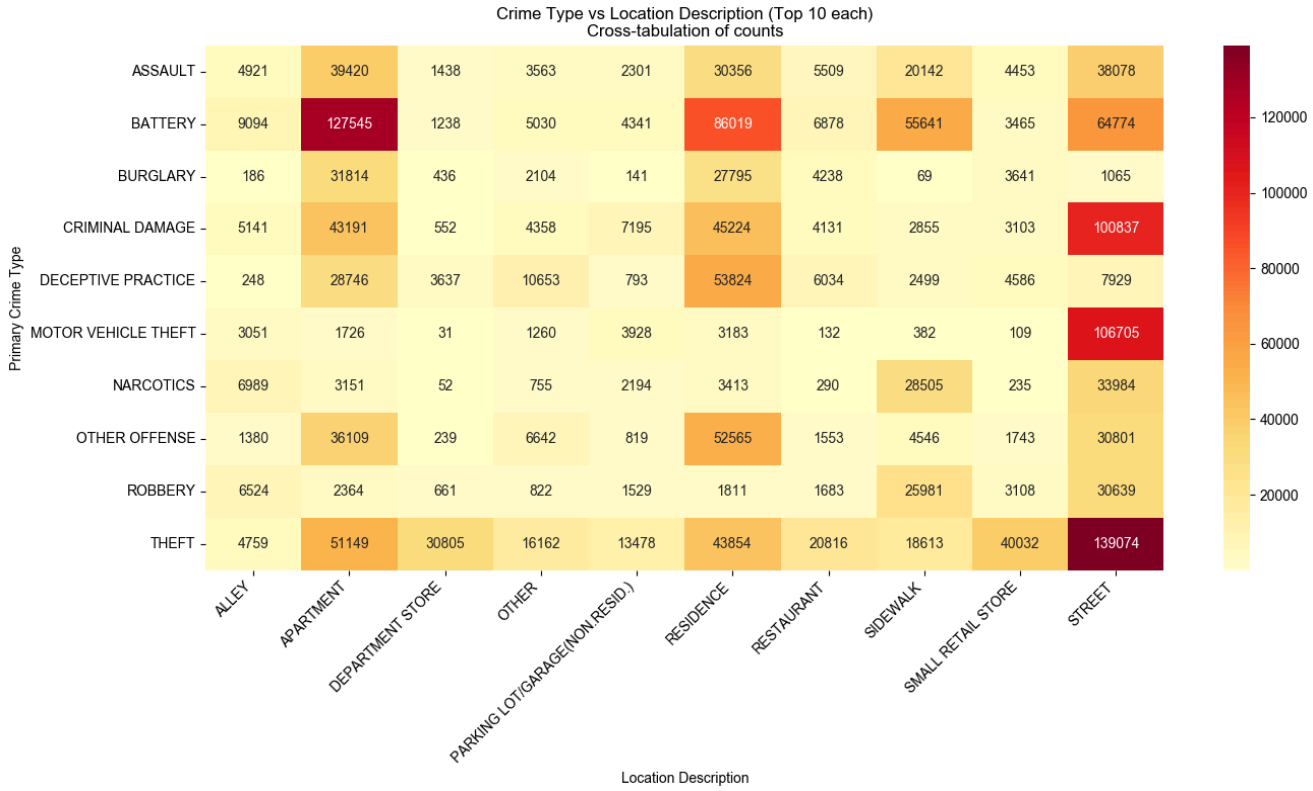


Figure 7

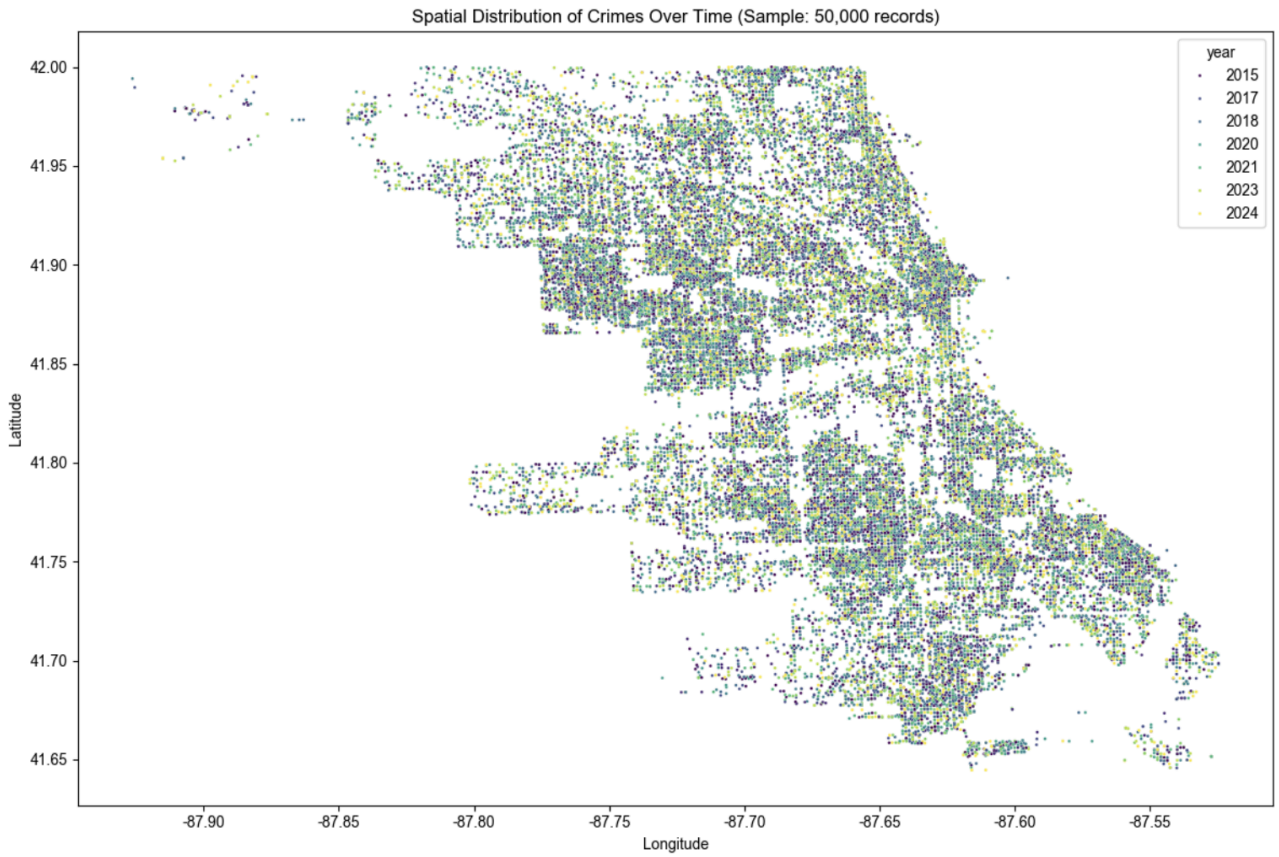


Figure 8